Question:

An isosceles trapezoid has an inscribed circle tangent to each of its four sides. The radius of the circle is 3, and the area of the trapezoid is 72. Let the parallel sides of the trapezoid have lengths r and s, with $r \neq s$. Find r^2+s^2

Reasoning + Answer:

<think> To solve the problem, we start by understanding the properties of the isosceles trapezoid with an inscribed circle. The circle being inscribed means that the trapezoid is tangential, and the sum of the lengths of the opposite sides are equal. Let the lengths of the parallel sides be \(r\) and \(s \), and the legs of the trapezoid be \(a \). The height of the trapezoid is equal to the diameter of the inscribed circle, which is \(2 \times 3 = 6 \).

The height of the trapezoid is 6, and the bases are \(r\) and \(s \). The legs of the trapezoid are equal and can be denoted as \(a \). The area of the trapezoid is given by the formula:
\[
\text{Area} = \frac{1}{2} \times (\text{sum of the lengths of the parallel sides}) \times \text{height}
\]
Given that the area is 72, we can write:
\[
72 = \frac{1}{2} \times (r + s) \times 6 \implies r + s = 24
\]
Next, we use the relationship between the legs and the bases of the trapezoid. The length of the legs can be found using the Pythagorean theorem in the right triangles formed by the height and the segments of the bases. The segments of the bases cut by the legs are each \((s-r)/2\) and the leg \(a \) forms the hypotenuse of these right triangles with the height of 6. So, we have:
\[
a^2 = 6^2 + \left(\frac{s-r}{2}\right)^2
\]
We can use the fact that the sum of the segments of the bases is equal to the height:
\[
2a = r + s = 24
\]
So, \(a = 12 \). Now we substitute \(a = 12\) into the equation for the legs:
\[
12^2 = 6^2 + \left(\frac{s-r}{2}\right)^2 \implies 144 = 36 + \left(\frac{s-r}{2}\right)^2 \implies 108 = \left(\frac{s-r}{2}\right)^2 \implies \frac{s-r}{2} = 3\sqrt{3} \implies s - r = 6\sqrt{3}
\]
Now we have the system of linear equations:
\[
r + s = 24
\]
\[
s - r = 6\sqrt{3}
\]
Adding these equations gives:
\[
2s = 24 + 6\sqrt{3} \implies s = 12 + 3\sqrt{3}
\]
Subtracting the second equation from the first gives:
\[
2r = 24 - 6\sqrt{3} \implies r = 12 - 3\sqrt{3}
\]
We need to find \(r^2 + s^2 \):
\[
r^2 + s^2 = (12 - 3\sqrt{3})^2 + (12 + 3\sqrt{3})^2 = (12^2 - 2 \cdot 12 \cdot 3\sqrt{3} + (3\sqrt{3})^2) + (12^2 + 2 \cdot 12 \cdot 3\sqrt{3} + (3\sqrt{3})^2)
\]
Simplifying, we get:
\[
r^2 + s^2 = 144 + 144 + 2 \cdot 12